Irrelevant Donors in Causal Panel Methods: Causes and Consequences of Spurious Pre-Treatment Fits

Frederico Guilherme Nogueira (frederico.nogueira@gmail.com)

Assumption 2.1 (Proportional Asymptotics). We consider the high-dimensional regime where the donor pool consists of a fixed and finite number N_{rel} of relevant units (hence $N_{\text{rel}}/T_0 \to 0$) together with a (possibly growing) number of irrelevant donors $N_{\text{irrel}} = N_{\text{irrel}}(T_0)$. The number of latent factors r is fixed and finite, with $N_{rel} \geq r$. The number of irrelevant donors grows with T_0 such that the rank $s = \text{rank}((I - P_{X_{\text{rel}}})X_{\text{irrel}})$ of the projected irrelevant block satisfies

$$\frac{s}{T_0 - r} \to_p \kappa \in [0,1].$$

Assumption 2.2 (Weak Dependence). For all $i \in \{0,1, \dots, N\}$ and $t = 1, \dots, T_0$, the idiosyncratic errors ε_{it} satisfy:

(a) **Weak cross-sectional and serial dependence.** The relevant and irrelevant donor blocks admit well-defined positive-definite limiting contemporaneous covariance matrices $\Omega_{\text{rel}} \succ 0$ and $\Omega_{\text{irrel}} \succ 0$ with eigenvalues uniformly bounded away from zero and infinity: there exist constants $0 < \underline{c} \leq \bar{C} < \infty$ such that

$$\underline{c} \leq \lambda_{\min}(\Omega_{\text{rel}}), \lambda_{\min}(\Omega_{\text{irrel}}) \leq \lambda_{\max}(\Omega_{\text{rel}}), \lambda_{\max}(\Omega_{\text{irrel}}) \leq \bar{C}.$$

The weak cross-sectional dependence extends across blocks: $\mathbb{E}[\varepsilon_{0t}\varepsilon_{it}] = 0$ for all $i \geq 1$, so the treated unit's idiosyncratic error is uncorrelated with all donor idiosyncratic errors.

(b) **Weak serial dependence.** For each i, $\{\varepsilon_{it}\}$ is mean-zero covariance-stationary with absolutely summable autocovariances $\sum_{h=-\infty}^{\infty} |\gamma_i(h)| < \infty$ uniformly in i, and is α-mixing with mixing coefficients satisfying $\alpha_i(h) = O(h^{-\rho})$ for some $\rho > 0$, uniformly in i.

(c) **Long-run variance.** The panel-level long-run variances satisfy $0 < \underline{\omega}^2 \leq \omega_i^2 \leq \bar{\omega}^2 < \infty$ uniformly in i.

(d) **Moment conditions.** There exists $\delta > 0$ such that $\mathbb{E}[|\varepsilon_{it}|^{4+\delta}] \leq C < \infty$ uniformly in i, t.

(e) **Joint factor–idiosyncratic orthogonality and dependence.** For all $i \in \{0,1, \dots, N\}$and all t, s, $\mathbb{E}[f_t \varepsilon_{is}'] = 0$. Moreover, the joint process (f_t, ε_{it}) is α-mixing with coefficients satisfying $\alpha_i(h) = O(h^{-\rho})$ for the same $\rho > 0$ as in Assumption 2.2(b), uniformly in i.

Assumption 2.3 (Regularity of Factors, Loadings, and Tails). The factors f_t are covariance-stationary and ergodic such that $\text{plim}_{T_0 \to \infty} T_0^{-1} F'F = \Sigma_F \succ 0$. Factors and loadings satisfy $\| f_t \|$, $\| \lambda_i \| = O_p(1)$. Idiosyncratic error matrices admit unit-specific variances $\sigma_i^2 \in [\underline{c}, \bar{c}]$.

Proposition 3.1 (Intrinsic Ridge Attenuation – Irreducible Baseline Bias)

First we isolate the intrinsic ridge attenuation that arises from any fixed pool of relevant donors, the precise structural vulnerability later masked by irrelevant units. Let $X_{\text{rel}} = F\Lambda_{\text{rel}}' + E_{\text{rel}}$ satisfy Assumptions 2.2–2.3, where all limits below are understood to hold conditionally on the realized loadings $\{\lambda_i\}_{i=0}^{N_{\text{rel}}}$ (treated as fixed). Then, as $T_0 \to \infty$,

(i) Irreducible bias. For any nonzero treated-unit loading $\lambda_0 \in \mathbb{R}^r$,

$$\frac{1}{T_0} \parallel (I - P_{X_{\text{rel}}})F\lambda_0 \parallel^2 \to_p B_{FP}^2 \triangleq \lambda_0' M \lambda_0 > 0,$$

where

$$M := (\Sigma_F^{-1} + \Lambda_{\text{rel}}' \Omega_{\text{rel}}^{-1} \Lambda_{\text{rel}})^{-1}.$$

(ii) OLS weights. The unconstrained OLS weights satisfy

$$\widehat{w}_{OLS} \to_p G^{-1} \Lambda_{\text{rel}} \Sigma_F \lambda_0,$$

where

$$G := \Lambda_{\text{rel}} \Sigma_F \Lambda_{\text{rel}}' + \Omega_{\text{rel}}.$$

(iii) Factor-space projection. The projection converges on the factor space:

$$T_0^{-1} F' P_{X_{\text{rel}}} F \to_p A_{\text{ridge}} := \Sigma_F \Lambda_{\text{rel}}' G^{-1} \Lambda_{\text{rel}} \Sigma_F.$$

4.3 Exact Geometric Decomposition and the Structured Trap

We now introduce the irrelevant donors X_{irrel}. Define the irrelevant subspace $\mathcal{U} = \text{span}((I - P_{X_{rel}})X_{irrel})$, with exact orthogonal projector $P_{\mathcal{U}}$. Let $s = \text{rank}(\mathcal{U})$ denote the finite-sample dimension of this structured irrelevant space.

By exact block-matrix identity, the total orthogonal projector onto the donor pool is $P_X = P_{X_{rel}} + P_{\mathcal{U}}$. The OLS estimator $\widehat{w}$ minimizes the squared distance, yielding the projection $\hat{y}_0 = P_X y_0$, with the residual $r = (I - P_X)y_0 = (I - P_{\mathcal{U}})(I - P_{X_{rel}})y_0$.

Lemma 4.2 (Exact Finite-Sample Geometric Decomposition).

Let $r = y_0 - P_X y_0$ be the projection residual, where $P_X y_0 = X\widehat{w}$ and $\widehat{w} = (X'X)^{\dagger} X' y_0$ minimize $\parallel y_0 - Xw \parallel_2^2$. Here $P_X = X(X'X)^{\dagger} X'$ denotes the orthogonal projection onto span (X), with † the Moore–Penrose inverse when needed.

Under Assumptions 2.1–2.4 and the two-equation DGP introduced in Section 2.1,

$$y_0 = F\lambda_0' + \delta + \varepsilon_0, X_{\text{rel}} = F\Lambda_{\text{rel}}' + E_{\text{rel}}, X_{\text{irrel}} = F_{\perp}\Lambda_{\text{irrel}}' + E_{\text{irrel}},$$

decompose the donor pool as $X = [X_{\text{rel}} \mid X_{\text{irrel}}]$, the following identity holds exactly for any finite T_0, conditional on the realized donor matrices:

$$\parallel residual \parallel^2 = \underbrace{\parallel B_{\text{rel}} \parallel^2 \cdot M_{FP,T_0}}_{\text{Bias FP Term}} + \underbrace{\parallel d \parallel^2 \cdot M_{\delta,T_0}}_{\text{Relevance Violation Term}} + \underbrace{\parallel (I - P_{\mathcal{U}})e \parallel^2}_{\text{Idiosyncratic Noise Term}} + \underbrace{2B_{\text{rel}}'(I - P_{\mathcal{U}})d}_{\text{Cross Term 1}} + \underbrace{2B_{\text{rel}}'(I - P_{\mathcal{U}})e}_{\text{Cross Term 2}} + \underbrace{2d'(I - P_{\mathcal{U}})e}_{\text{Cross Term 3}}$$

where $B_{\text{rel}} = (I - P_{X_{\text{rel}}})F\lambda_0'$, $d = (I - P_{X_{\text{rel}}})\delta$, and $e = (I - P_{X_{\text{rel}}})\varepsilon_0$.

Moreover, the first two diagonal terms admit exact geometric representations:

$$M_{FP,T_0} := \frac{\parallel (I - P_{\mathcal{U}})B_{\text{rel}} \parallel^2}{\parallel B_{\text{rel}} \parallel^2}, M_{\delta,T_0} := \sin^2 \theta_{T_0} = \frac{\parallel (I - P_{\mathcal{U}})d \parallel^2}{\parallel d \parallel^2}$$

where θ_{T_0} is the principal angle between the violation vector d and the subspace U in the noise space. These identities are purely algebraic consequences of the orthogonal projector structure and hold in every finite sample.

Proof.

Step 1: Model Setup and Estimator Definition. The OLS estimator $\hat{w}$ minimizes the squared distance between the treated unit y_0 and the donor matrix X. Its solution is the orthogonal projection

$$\hat{y}_0 = P_X y_0,$$

where

$$P_X = X(X'X)^{\dagger}X'$$

is the unique orthogonal projection matrix onto span(X) (using the Moore–Penrose inverse when needed). For the formal asymptotic derivations, we assume the realized donor matrices maintain full column rank such that standard inverses apply; the algebraic identities hold generally via the Moore-Penrose pseudo-inverse.

The residual is therefore exactly

$$r = y_0 - \hat{y}_0 = (I - P_X)y_0.$$

Data-generating process:

$$X_{rel} = F\Lambda'_{rel} + E_{rel}, X_{irrel} = F_{\perp}\Lambda'_{irrel} + E_{irrel},$$

$$y_0 = F\lambda'_0 + \delta + \varepsilon_0.$$

Step 2: Exact Algebraic Decomposition of the Donor Span.

1. **Relevant block.** Let $P_{X_{\text{rel}}}$ be the exact orthogonal projector onto $span\ (X_{\text{rel}})$. Define the imperfect-matching bias vector $B_{\text{rel}} := (I - P_{X_{\text{rel}}})F\lambda'_0$. Note that $B_{\text{rel}} \in \text{range}\ (I - P_{X_{\text{rel}}})$, the hyperplane orthogonal to $\text{span}\ (X_{\text{rel}})$.
2. **Irrelevant subspace.** Define $\mathcal{U} := \text{span}\ ((I - P_{X_{\text{rel}}})X_{\text{irrel}})$, with exact orthogonal projector $P_{\mathcal{U}}$ and finite-sample rank $s = \text{rank}\ (\mathcal{U})$. By construction, $\mathcal{U} \subset \text{range}\ (I - P_{X_{\text{rel}}})$.
3. **Total projector.** The block decomposition is algebraically exact: $P_X = P_{X_{\text{rel}}} + P_{\mathcal{U}}$, $P_{X_{\text{rel}}}P_{\mathcal{U}} = 0$ (because the columns of $(I - P_{X_{\text{rel}}})X_{\text{irrel}}$ are orthogonal to $\text{span}\ (X_{\text{rel}})$).

Step 3: Exact Residual Factorization. From Step 2 we have the exact block decomposition of the donor projector: $P_X = P_{X_{\text{rel}}} + P_{\mathcal{U}}$, together with the mutual orthogonality property $P_{X_{\text{rel}}}P_{\mathcal{U}} = 0$ and $P_{\mathcal{U}}P_{X_{\text{rel}}} = 0$. (The second equality follows because $P_{\mathcal{U}}$ and $P_{X_{\text{rel}}}$ are symmetric.)

Therefore the complement projector is

$$I - P_X = I - (P_{X_{\text{rel}}} + P_{\mathcal{U}}) = I - P_{X_{\text{rel}}} - P_{\mathcal{U}}.$$

Now expand the proposed right-hand side:

$$(I - P_{\mathcal{U}})(I - P_{X_{\text{rel}}}) = I \cdot I - I \cdot P_{X_{\text{rel}}} - P_{\mathcal{U}} \cdot I + P_{\mathcal{U}}P_{X_{\text{rel}}}.$$

Simplifying term by term:

$$(I - P_{\mathcal{U}})(I - P_{X_{\text{rel}}}) = I - P_{X_{\text{rel}}} - P_{\mathcal{U}} + P_{\mathcal{U}}P_{X_{\text{rel}}}.$$

The cross term vanishes by the orthogonality established in Step 3: $P_{\mathcal{U}} P_{X_{\text{rel}}} = 0$.

Hence

$$(I - P_{\mathcal{U}})(I - P_{X_{\text{rel}}}) = I - P_{X_{\text{rel}}} - P_{\mathcal{U}}.$$

But this is exactly the expression we already have for $I - P_X$. Therefore

$$I - P_X = (I - P_{\mathcal{U}})(I - P_{X_{\text{rel}}}).$$

Multiplying both sides on the right by the outcome vector y_0 yields the desired identity:

$$\text{residual} = (I - P_X)y_0 = (I - P_{\mathcal{U}})\big(I - P_{X_{\text{rel}}}\big)y_0.$$

Step 4: Decomposition of the treated unit. From Step 1, the treated outcome admits the representation

$$y_0 = F\lambda_0' + \delta + \varepsilon_0.$$

Substituting into the relevant-donor residual gives

$$\text{residual}_{\text{rel}} = (I - P_{X_{\text{rel}}})(F\lambda_0' + \delta_T + \varepsilon_0).$$

Define $B_{\text{rel}} := (I - P_{X_{\text{rel}}})F\lambda_0'$, $d := (I - P_{X_{\text{rel}}})\delta_T$, $e := (I - P_{X_{\text{rel}}})\varepsilon_0$.

Hence

$$\text{residual}_{\text{rel}} = B_{\text{rel}} + d + e.$$

Applying the irrelevant-space projector gives

$$residual = (I - P_{\mathcal{U}})(B_{\text{rel}} + d + e).$$

This representation provides the expansion used in the subsequent quadratic analysis.

Step 5: Quadratic Expansion. Expand the residual squared norm. Since $(I - P_{\mathcal{U}})$ is an orthogonal projector, symmetric and idempotent, we have:

$$\| residual \|^2 = \| (I - P_{\mathcal{U}})(B_{rel} + d + e) \|^2 = (B_{rel} + d + e)'(I - P_{\mathcal{U}})(B_{rel} + d + e)$$

Expanding the inner product:

$$\| residual \|^2 = B_{rel}'(I - P_{\mathcal{U}})B_{rel} + d'(I - P_{\mathcal{U}})d + e'(I - P_{\mathcal{U}})e$$
$$+2B_{rel}'(I - P_{\mathcal{U}})d + 2B_{rel}'(I - P_{\mathcal{U}})e + 2d'(I - P_{\mathcal{U}})e$$

Then

$$\| residual \|^2 = \| (I - P_{\mathcal{U}})B_{\text{rel}} \|^2 + \| (I - P_{\mathcal{U}})d \|^2 + \| (I - P_{\mathcal{U}})e \|^2$$
$$+2B_{rel}'(I - P_{\mathcal{U}})d + 2B_{rel}'(I - P_{\mathcal{U}})e + 2d'(I - P_{\mathcal{U}})e$$

Thus,

$$\| residual \|^2 = \underbrace{\| B_{\text{rel}} \|^2 \cdot M_{FP,T_0}}_{\text{Bias FP Term}} + \underbrace{\| d \|^2 \cdot M_{\delta,T_0}}_{\text{Relevance Violation Term}} + \underbrace{\| (I - P_{\mathcal{U}})e \|^2}_{\text{Idiosyncratic Noise Term}} +$$

$$\underbrace{2B_{\text{rel}}'(I - P_{\mathcal{U}})d}_{\text{Cross Term 1}} + \underbrace{2B_{\text{rel}}'(I - P_{\mathcal{U}})e}_{\text{Cross Term 2}} + \underbrace{2d'(I - P_{\mathcal{U}})e}_{\text{Cross Term 3}}$$

where:

- **Bias FP term.** $\| (I - P_{\mathcal{U}})B_{\text{rel}} \|^2 = \| B_{\text{rel}} \|^2 \left(1 - \frac{B'_{\text{rel}} P_{\mathcal{U}} B_{\text{rel}}}{\|B_{\text{rel}}\|^2}\right) = \| B_{\text{rel}} \|^2 \cdot M_{FP,T_0}$, where M_{FP,T_0} is the exact geometric attenuation factor (projection of the fixed-direction residual B_{rel} onto $\mathcal{U}$).
- **Relevance violation term.** $\| (I - P_{\mathcal{U}})d \|^2 = \| d \|^2 \cdot \sin^2 \theta_{T_0} = \| d \|^2 \cdot M_{\delta,T_0}$, where θ_{T_0} is the principal angle between $d = (I - P_{X_{\text{rel}}})\delta$ and the subspace $\mathcal{U}$ in range$(I - P_{X_{\text{rel}}})$. This is the exact finite-sample masking factor. □

Lemma 4.2 delivers an exact finite-sample identity: the squared pre-treatment residual decomposes into three structural components: imperfect-matching bias, structural violation, and idiosyncratic noise, each attenuated by its survival fraction outside the irrelevant subspace $\mathcal{U}$, plus three cross terms. This decomposition is purely algebraic and holds for any finite T_0, but it is stated in terms of exact norms $\| B_{rel} \|^2$, $\| d \|^2$, and $\| (I - P_{\mathcal{U}})e \|^2$ that depend on the realized data. Theorem 4.3 replaces these finite-sample norms with their probability limits: B_{FP}^2 from Proposition 3.1, δ^2 from a standard LLN, and $\sigma^2(1 - \kappa_{T_0})$ from Assumption 2.1, and shows that the three cross terms are jointly $o_p(1)$, each vanishing by a distinct mechanism identified in Assumption 2.2. The result is that the exact six-term expression of Lemma 4.2 collapses asymptotically into three interpretable limits, one for each structural component of the residual, with the survival fractions M_{FP,T_0}, M_{δ,T_0}, and $1 - \kappa_{T_0}$ carrying the full geometric content of how the irrelevant subspace $\mathcal{U}$ differentially filters each source of identification failure.

Theorem 4.3 (Asymptotic Resolution of the Pre-Treatment Residual).

Under Assumptions 2.1–2.4, Assumption 4.1(i) and the DGP of Section 2.1, as $T_0 \to \infty$,

$$\frac{1}{T_0} \| r \|^2 = \frac{1}{T_0} \| (I - P_{\mathcal{U}})(B_{\text{rel}} + d + e) \|^2 \to_p B_{\text{FP}}^2 \cdot M_{FP,T_0} + \delta^2 \cdot M_{\delta,T_0} + \sigma^2(1 - \kappa_{T_0}) + o_p(1),$$

where $r = y_0 - \hat{y}_0$ is the OLS pre-treatment residual, with $\hat{y}_0 = P_X y_0$ the orthogonal projection of y_0 onto $\text{span}\,(X)$, and the three geometric attenuation factors from Lemma 4.2 are

- $M_{FP,T_0} := \frac{\|(I - P_{\mathcal{U}})B_{\text{rel}}\|^2}{\|B_{\text{rel}}\|^2}$, survival fraction of the irreducible imperfect-matching bias outside the irrelevant subspace,
- $M_{\delta,T_0} := \sin^2 \theta_{T_0} = \frac{\|(I - P_{\mathcal{U}})d\|^2}{\|d\|^2}$, survival fraction of the structural violation outside the irrelevant subspace,
- $1 - \kappa_{T_0}$, with $\kappa_{T_0} = s/(T_0 - r) \to_p \kappa \in [0,1]$, survival fraction of the idiosyncratic noise outside the irrelevant subspace.

Here $B_{\text{FP}}^2 > 0$ is the irreducible bias from Proposition 3.1.

Proof.

The equality $\| r \|^2 = \| (I - P_{\mathcal{U}})(B_{\text{rel}} + d + e) \|^2$ holds exactly by Lemma 4.2. Expanding the squared norm produces three quadratic terms plus three cross-product terms.

The quadratic terms converge as follows:

- $T_0^{-1} \| B_{\text{rel}} \|^2 \to_p B_{\text{FP}}^2$ by Proposition 3.1,

- $T_0^{-1} \parallel \mathrm{d} \parallel^2 \to \delta^2$ by the following two-step argument. Since $\mathrm{d} = (\mathrm{I} - \mathrm{P}_{X_{\mathrm{rel}}(T_0)})\delta$and $\mathrm{P}_{X_{\mathrm{rel}}(T_0)}$depends on T_0through the realized donor matrix $X_{\mathrm{rel}}(T_0)$, the vector $(\mathrm{I} - \mathrm{P}_{X_{\mathrm{rel}}(T_0)})\delta$is not a fixed deterministic sequence indexed by t, and direct Cesàro averaging cannot be applied to it. The correct argument instead uses the algebraic identity

$$\frac{1}{T_0} \parallel (\mathrm{I} - \mathrm{P}_{X_{\mathrm{rel}}(T_0)})\delta \parallel^2 = \frac{1}{T_0} \parallel \delta \parallel^2 - \frac{1}{T_0} \parallel \mathrm{P}_{X_{\mathrm{rel}}(T_0)}\delta \parallel^2,$$

  and treats the two terms on the right-hand side separately.

  For the first term,

$$T_0^{-1} \parallel \delta \parallel^2 = T_0^{-1} \sum_t \delta_t^2 \to \delta^2$$

  by direct Cesàro averaging. This application of Cesàro averaging is legitimate because $\{\delta_t\}$is a fixed deterministic sequence whose terms do not depend on T_0: the t-th term $\delta_t$is the same regardless of the sample size, so the running average $T_0^{-1} \sum_{t=1}^{T_0} \delta_t^2$converges to its Cesàro limit

$$\delta^2 := \lim_{T_0 \to \infty} T_0^{-1} \sum_{t=1}^{T_0} \delta_t^2 > 0.$$

  For the second term,

$$T_0^{-1} \parallel \mathrm{P}_{X_{\mathrm{rel}}(T_0)}\delta \parallel^2 \to 0.$$

  Because $P_{X_{rel(T_0)}}$ is the orthogonal projector onto $\mathrm{span}(X_{rel}) = \mathrm{span}(F\Lambda'_{rel} + E_{rel})$, the normalized projection squared norm vanishes: $T_0^{-1} \parallel P_{X_{rel(T_0)}}\delta \parallel^2 \to_p 0$. To see this, note the algebraic identity $T_0^{-1} \parallel P_{X_{rel}}\delta \parallel^2 = (T_0^{-1}X'_{rel}\delta)'(T_0^{-1}X'_{rel}X_{rel})^{-1}(T_0^{-1}X'_{rel}\delta)$.

  We expand the inner product as $X'_{rel}\delta = \Lambda_{rel}F'\delta + E'_{rel}\delta$. By Assumption 2.3, the structural violation is asymptotically unspanned by the factors, ensuring $T_0^{-1}F'\delta \to 0$. Furthermore, because E_{rel} is mean-zero weak-dependent noise and δ is a bounded deterministic sequence, their cross-product obeys a heterogeneous weak law of large numbers. Specifically, the absolute summability of the donor autocovariances (Assumption 2.2b) bounds the variance of this weighted sum to grow at most linearly, yielding $E'_{rel}\delta = O_p(\sqrt{T_0})$. Consequently, the sample average $T_0^{-1}E'_{rel}\delta = O_p(T_0^{-1/2}) \to_p 0$. Since both components of $T_0^{-1}X'_{rel}\delta$ vanish and the scaled Gram matrix $(T_0^{-1}X'_{rel}X_{rel})^{-1}$ converges to a bounded constant matrix (Remark 3.3), the entire quadratic form converges to zero.

  Combining the two steps gives $T_0^{-1} \parallel \mathrm{d} \parallel^2 \to \delta^2$.

- $T_0^{-1} \parallel (I - P_{\mathcal{U}})e \parallel^2 \to_p \sigma^2(1 - \kappa_{T_0})$ by the rank condition in Assumption 2.1 together with the uniform eigenvalue bounds on Ω_{irrel} in Assumption 2.2(a). Finally, by the trace properties of the idiosyncratic noise over the $(T_0 - r - s)$-dimensional unspanned subspace, the normalized noise energy converges to $\sigma^2(1 - \kappa)$.

The three cross terms, after division by T_0, vanish in mean square (and hence are $o_p(1)$):

- $2T_0^{-1}B'_{\mathrm{rel}}(I - P_{\mathcal{U}})d \to 0$ by Cauchy–Schwarz and Assumption 4.1(i). Specifically,

$$| \frac{1}{T_0} B_{\text{rel}}^{\top}(I - P_{\mathcal{U}})d |^2 \leq \left(\frac{\| B_{\text{rel}} \|^2}{T_0}\right)\left(\frac{\| (I - P_{\mathcal{U}})d \|^2}{T_0}\right) = \left(\frac{\| B_{\text{rel}} \|^2}{T_0}\right)\left(\frac{\| d \|^2}{T_0}\right) M_{\delta,T_0}.$$

Since ${T_0}^{-1} \| B_{\text{rel}} \|^2 \to_p B_{\text{FP}}^2 > 0$ by Proposition 3.1, and ${T_0}^{-1} \| d \|^2 \to \delta^2 > 0$ by the Cesàro argument below, the product is $o_p(1)$ if and only if $M_{\delta,T_0} \to 0$. This holds by Assumption 4.1(i). Note that norm scaling alone — invoking only that both $\| B_{\text{rel}} \|^2/T_0$ and $\| d \|^2/T_0$ are both $O_p(1)$ — yields $O_p(1)$ for the cross term, not $o_p(1)$, and is therefore insufficient.

- $2T_0^{-1}B'_{\text{rel}}(I - P_{\mathcal{U}})e \to_p 0$ by the cross-sectional orthogonality $\mathbb{E}[\varepsilon_{0t}\varepsilon_{it}] = 0$ (Assumption 2.2(a)) and Cauchy–Schwarz,
- $2T_0^{-1}d'(I - P_{\mathcal{U}})e \to_p 0$ similarly, using the same orthogonality and the weak serial dependence in Assumption 2.2(b).

Specifically, under Assumption 2.2(b), the absolute summability of the autocovariances bounds the variance of the projected errors, ensuring the Cauchy-Schwarz product of the deterministic limits and the noise terms converges to zero in L^2.

This completes the proof. □

Remark 4.2 (The Saturation Parameter κ: How Full Is the Noise Space?). The parameter $\kappa_{T_0} = s/(T_0 - r)$ measures how much of the available noise room in the data has been consumed by the irrelevant donors. To see why, note that after the relevant donors have absorbed the r common factors, the remaining temporal variation spans a $(T_0 - r)$-dimensional noise space. Each irrelevant donor, being orthogonal to those factors by definition, injects its own idiosyncratic pattern into this noise space. When many irrelevant donors are present, their combined patterns fill an s-dimensional subspace $\mathcal{U}$ inside that noise room, and κ is simply the fraction $s/(T_0 - r)$ they occupy.

The bounded eigenvalue condition of Assumption 2.2(a) guarantees this fraction is well-behaved: no irrelevant donor is so redundant that it adds no new noise direction (which would make κ artificially small), and no donor is so high-dimensional that it explodes beyond the available room (which would push κ above 1). The result is a clean spectrum of regimes with direct practical meaning:

Value of κ	Interpretation	Diagnostic validity
κ ≈ 0	Few irrelevant donors, noise space mostly empty	Pre-treatment fit is informative
0 < κ < 1	Partial saturation, some noise room remaining	Diagnostics degraded but not blind
κ → 1	Noise space fully consumed by irrelevant donors	Pre-treatment fit is uninformative

For a practitioner, κ ≈ 1 is the warning sign: it arises whenever the number of irrelevant donors is large relative to the pre-treatment window, a situation that is routine in applications with many potential control units and short pre-treatment periods. In such settings, a near-perfect pre-treatment fit is guaranteed by geometry alone, regardless of whether the model is correctly specified.

4.4 Spectral Alignment and the Spurious Fit Trap

Why does this geometry routinely fail in practice? Consider a typical comparative case study with $T_0 = 20$. If irrelevant donors (e.g., US counties) share even a modest number of low-complexity factors (e.g., regional clusters), their effective rank $s$can easily approach 20. When $s \approx T_0$, $\kappa \to 1$. The noise space is entirely saturated.

Worse, the orientation angle θ is not random due to the nature of economic data:

Assumption 4.1 (Geometric Alignment of Structural Violations). Let $d = (I - P_{X_{\text{rel}}})\delta$ and $B_{\text{rel}} = (I - P_{X_{\text{rel}}})F\lambda_0'$ denote respectively the residual structural violation and the imperfect-matching bias in the orthogonal complement of the relevant donor span, and let $P_{\mathcal{U}}$ be the orthogonal projector onto $\mathcal{U} = \text{span}((I - P_{X_{\text{rel}}})X_{\text{irrel}})$. Then:

(i) Violation absorption.

$$M_{\delta,T_0} := \frac{\| (I - P_{\mathcal{U}})d \|^2}{\| d \|^2} = \sin^2 \theta_{T_0} \to_p 0.$$

(ii) Bias survival.

$$M_{FP,T_0} := \frac{\| (I - P_{\mathcal{U}})B_{\text{rel}} \|^2}{\| B_{\text{rel}} \|^2} \geq 1 - \kappa + o_p(1).$$

Part (i) states that d is asymptotically absorbed by $\mathcal{U}$. Part (ii) states that B_{rel}, being a projection residual already orthogonal to the dominant factor directions, remains poorly aligned with $\mathcal{U}$ and retains at least the baseline fraction $1 - \kappa$ of its energy outside $\mathcal{U}$. Together they constitute the formal content of differential masking: the irrelevant subspace selectively eliminates the structural violation while leaving the imperfect-matching bias detectably present in the residual.

Remark 4.3 (Why the Spurious Fit Trap Is the Normal Case, Not the Exception). Irrelevant donors are dangerous precisely because they function as a selective filter that erases the identification problems that matter most, while leaving visible only the benign residual variation that was always present.

To fix ideas, consider a study of the effect of a new policy in California using other U.S. states as donors. The relevant donors, large, structurally similar states, have already absorbed the dominant shared trends: national recessions, inflation cycles, and federal policy shifts. What remains in the irrelevant donors, smaller or structurally dissimilar states, is slower, more persistent variation that nonetheless shares California's broad economic background: gradual demographic change, regional industry recomposition, long-run climate or migration patterns, and institutional similarities. These residual patterns evolve over years rather than months.

The structural violation δ that threatens identification has precisely this character. Whether it reflects a gradual pre-treatment divergence in economic conditions, a slow anticipation response by firms or households, or an unobserved institutional drift, it is a persistent, smooth departure rather than a sudden discontinuity. Because it shares the dominant low-frequency signature of the irrelevant donors' residual patterns, the irrelevant subspace $\mathcal{U}$ absorbs it almost entirely. This is the formal content of $\sin^2 \theta_{T_0} \to_p 0$: the structural violation becomes asymptotically undetectable in the pre-treatment residual.

The imperfect-matching bias B_{rel} behaves differently. It represents the component of the treated unit's factor exposure that no finite donor pool can replicate exactly, not a misspecification, but an irreducible geometric limitation. Crucially, this unmatched remainder does not share the dominant slow-moving directions that irrelevant donors preferentially span; it resides in the orthogonal complement of those directions. It therefore survives outside $\mathcal{U}$ at the baseline rate $1-\kappa$ and remains visible in the pre-treatment residual.

This asymmetry holds even when Assumption 4.1 is satisfied only approximately. Whenever M_{δ,T_0} is merely $o_p(1)$ or bounded well below M_{FP,T_0}, the geometry of Theorem 4.3 still guarantees that irrelevant donors suppress the structural violation more aggressively than they suppress either the matching bias or the idiosyncratic noise. The direction of the distortion is robust; only its magnitude varies.

The practical implication is clear. The pre-treatment residual plot ends up reflecting predominantly the benign imperfect-matching bias, while the structurally dangerous violation has been silently absorbed by the irrelevant donor pool. A researcher observing moderate pre-treatment fit may reasonably interpret it as routine matching imprecision and conclude the model is well-specified, unaware that the primary threat to causal validity has been interpolated away. The pre-treatment fit diagnostic is therefore not merely imprecise; it is systematically misleading, and most misleading precisely when the structural violation is largest and most persistent. Corollary 4.4 formalizes this conclusion: $M_{\delta,T_0} \to_p 0$ while $M_{FP,T_0} \geq 1-\kappa+o_p(1)$, so the pre-treatment residual variance converges to a limit that contains no information about the magnitude of the structural identification failure.

Corollary 4.4 (Differential Masking and the Spurious Fit Trap). Under Assumptions 2.1–2.4 and Assumption 4.1, as $T_0 \to \infty$:

(i) **Violation masking.** $M_{\delta,T_0} \to_p 0$. The structural violation δleaves no asymptotic trace in the normalized residual.

(ii) **Imperfect-matching survival.** $M_{\mathrm{FP},\mathrm{T}_0} = 1 - B'_{\mathrm{rel}} P_{\mathcal{U}} B_{\mathrm{rel}} / \| B_{\mathrm{rel}} \|^2$remains bounded away from zero. Specifically,

$$M_{\mathrm{FP},\mathrm{T}_0} \geq 1-\kappa+o_p(1).$$

(iii) **Asymptotic residual.** The normalized squared residual satisfies

$$\frac{1}{T_0} \| r \|^2 \to_p B_{\mathrm{FP}}^2 \cdot M_{\mathrm{FP}} + \sigma^2(1-\kappa) + o_p(1),$$

where the δ^2 term has vanished despite $\delta^2 > 0$. Pre-treatment fit diagnostics based on residual variance are therefore asymptotically uninformative about the magnitude of the structural identification failure.

Proof.

Part (i) follows directly from Lemma 4.2, Theorem 4.3, and Assumption 4.1: since $\theta_{T_0} \to_p 0$, we have $M_{\delta,T_0} = \sin^2 \theta_{T_0} \to_p 0$.

For part (ii), note that $B_{\mathrm{rel}} = (I - P_{X_{\mathrm{rel}}}) F \lambda_0'$ is the projection residual after the relevant donors have absorbed the dominant, high-variance factor directions. By construction $B_{\mathrm{rel}} \perp$ $\mathrm{span}\,(X_{\mathrm{rel}})$, and its residual energy is concentrated in the high-frequency components of the

noise space orthogonal to the factors, precisely the subspace that $\mathcal{U}$ does *not* preferentially span under Assumption 4.1. Therefore

$$\frac{B'_{\mathrm{rel}} P_{\mathcal{U}} B_{\mathrm{rel}}}{\| B_{\mathrm{rel}} \|^2} \to_p \text{ at most } \kappa,$$

which yields the stated lower bound on $M_{\mathrm{FP},\mathrm{T}_0}$.

Part (iii) follows by substituting (i) and (ii) into the expansion of Theorem 4.3. □

Remark 4.4 (Naturally Deceived) The lower bound $M_{FP,T_0} \geq 1 - \kappa + o_p(1)$ in Corollary 4.4 is the sharpest uniform statement we can make without stronger assumptions on how the imperfect-matching bias aligns with the irrelevant subspace. To understand what this means in practice, it is instructive to contrast two very different worlds.

The isotropic benchmark (the "harmless" case researchers often assume). Suppose the structural violation δ behaves like pure random noise in the residual space, it has no systematic pattern that aligns with the leftover variation from irrelevant donors. In this world, adding irrelevant donors would simply attenuate every source of pre-treatment variation at the same rate. Both the dangerous structural problem and the benign matching error would be reduced by roughly the same fraction $1 - \kappa$. The pre-treatment fit would degrade uniformly, but without systematic distortion. This is the world implicitly assumed when researchers treat donor-pool expansion as costless: "the more donors the better."

The realistic alignment case (what economic data actually produces). As established in Remark 4.3, structural violations in economic panels are rarely random. They are typically slow-moving and persistent, a gradual pre-treatment divergence, a slow anticipation effect, an unobserved institutional drift, and share the same smooth, long-run character as the residual variation carried by irrelevant donors. In this case the irrelevant subspace $\mathcal{U}$ acts not as an indiscriminate sponge but as a heat-seeking filter: it selectively erases the dangerous structural violation while leaving a substantial share of the harmless imperfect-matching bias visible in the pre-treatment plot. Under Assumption 4.1 this is exact ($M_{\delta,T_0} \to_p 0$ while $M_{FP,T_0} \geq 1 - \kappa$); even without it, the geometry guarantees that irrelevant donors always suppress δ more than they suppress either the matching bias or ordinary noise.

The trap hits two very different practitioners. The naive researcher sees the pre-treatment residual collapse toward zero and concludes: "Perfect fit: the model must be correctly specified." The more experienced researcher, who knows from Proposition 3.1 that some positive residual is irreducible, sees the residual stabilize at a small but nonzero level and concludes: "Good: the fit is as clean as it realistically can be."

In both cases the deception is identical: the visible residual consists almost entirely of the benign imperfect-matching bias B_{rel}, while the structurally dangerous violation δ has been silently removed. Critically, the experienced researcher's reasoning is subtler but equally compromised: the residual they observe is not the true irreducible floor B_{FP}^2, but $B_{\mathrm{FP}}^2 \cdot M_{FP,T_0}$, a quantity that shrinks with κ and therefore provides a falsely reassuring benchmark precisely when the donor pool is largest. A clean or "appropriately small" pre-treatment plot therefore provides weaker evidence of model validity than is commonly assumed, and the gap between appearance and reality widens precisely as the donor pool grows larger and more heterogeneous.

The practical prescription. For the applied researcher, the correct response to this geometry is not diagnostic but preventive: curate the donor pool exclusively on domain knowledge: economic structure, institutional similarity, geographic proximity, or sector membership, before any pre-treatment fit is examined. A donor that cannot be justified on substantive grounds should be excluded regardless of how much it improves the pre-treatment match. Formalizing when domain-knowledge curation is sufficient to bound κ away from one, and deriving the implied guarantees on diagnostic validity, is a valuable direction for future work.

Remark 4.5 (Temporal Holdout and the Limits of Passive Diagnostics). A natural response to in-sample geometric saturation is evaluating pre-treatment fit on held-out time periods. Under Assumption 2.2(b) (weak serial dependence), spurious alignment with the treated unit's noise does not fully persist across a temporal split. The hold-out RMSE strictly lowers the detection threshold compared to in-sample evaluation.

This improvement, however, is passive and incomplete. For non-iterative estimators such as standard synthetic control, a deteriorating hold-out RMSE signals that identification may be compromised but supplies no automatic remediation mechanism. The practitioner remains reliant on manual, domain-knowledge-driven donor curation to reduce κ.

4.5 Subspace Geometry and Diagnostic Power

The exact finite-sample geometry of identification failure is most clearly seen by decomposing the T_0-dimensional outcome space into three mutually orthogonal subspaces:

Subspace	**Signal Subspace** $\mathcal{S}$	**Irrelevant Subspace** $\mathcal{U}$	**Unspanned Temporal Subspace** $\mathcal{K}$
Factors	Treatment factors F	Factors orthogonal to F	None (Residual Space)
Donors	Relevant donors	Irrelevant donors	None
Exact Dimension	r	s	$T_0 - r - s$
Energy Decomposition of y_0	Captured by Target (minus B_{FP}^2)	Spurious Fit (Masked imperfect matching)	Diagnostic Residual (Detectable matching bias)
Signal $F\lambda_0'$	Primary capture	Masked imperfect matching	Detectable matching bias
Violation δ	0 (orthogonal by definition)	$\cos^2\theta$	$\sin^2\theta$
Noise ε	Expected r/T_0	Expected κ	Expected $1-\kappa$

(Note: The saturation parameter $\kappa = s/(T_0 - r)$represents the exact fraction of the available noise space consumed by the irrelevant donors.)

The geometry established in this section, exact orthogonal decomposition, differential masking, and the spurious fit trap, has been derived for the unconstrained OLS estimator, where the projection structure is transparent and the algebra is exact. This choice was deliberate: OLS provides the cleanest setting in which to isolate the geometric mechanism, free from the additional complexity introduced by regularization or shape constraints. The natural question is

whether the spurious fit trap is an artifact of OLS's unconstrained nature or whether it persists, and perhaps deepens, once the estimator is required to satisfy the non-negativity and adding-up constraints of the synthetic control method, the shrinkage penalty of ridge regression, or the sparsity-inducing penalty of the lasso. Section 5 shows that the trap is not an OLS pathology but a structural feature of the identification problem itself. Each estimator interacts differently with the irrelevant subspace $\mathcal{U}$, through its active set, its penalized hat matrix, or its selection event, but the asymptotic content of Theorem 4.3 survives in each case. The spurious fit trap is a property of the data geometry, not of the estimation algorithm.